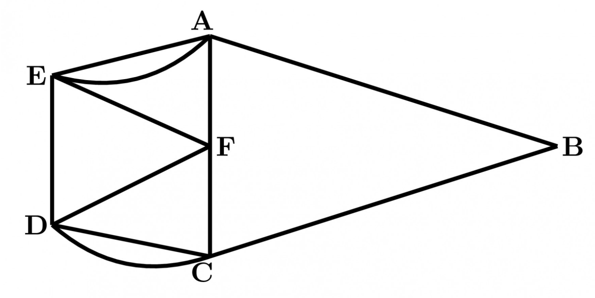
9 Discrete Mathematics: Graph Theory (34)

9.0.1 NIELIT 2017 July Scientist B (CS) - Section B: 13

For the graph shown, which of the following paths is a Hamilton circuit?



A. $ABCDCFDEFAEA$
B. $AEDCBAF$
C. $AEFDCBA$
D. $AFCDEBA$

nielit2017july-scientistb-cs discrete-mathematics graph-theory

Answer key☟

9.0.2 NIELIT 2017 July Scientist B (IT) - Section B: 15

The number of diagonals that can be drawn by joining the vertices of an octagon is

A. 28 B. 48 C. 20 D. None of the option

nielit2017july-scientistb-it discrete-mathematics graph-theory

Answer key☟

9.0.3 NIELIT 2017 July Scientist B (IT) - Section B: 3

A path in graph G, which contains every vertex of G and only once?

A. Euler circuit
B. Hamiltonian path
C. Euler Path
D. Hamiltonian Circuit

nielit2017july-scientistb-it discrete-mathematics graph-theory

Answer key☟

9.0.4 NIELIT 2017 July Scientist B (IT) - Section B: 2

Which of the following is an advantage of adjacency list representation over adjacency matrix representation of a graph?

A. In adjacency list representation, space is saved for sparse graphs.
B. Deleting a vertex in adjacency list representation is easier than adjacency matrix

representation.

C. Adding a vertex in adjacency list representation is easier than adjacency matrix representation.

D. All of the option.

nielit2017july-scientistb-it discrete-mathematics graph-theory

Answer key

9.0.5 NIELIT 2016 MAR Scientist B - Section B: 1

The number of ways to cut a six sided convex polygon whose vertices are labeled into four triangles using diagonal lines that do not cross is

A. 13 B. 14 C. 12 D. 11

nielit2016mar-scientistb discrete-mathematics graph-theory

Answer key

9.0.6 NIELIT 2017 DEC Scientific Assistant A - Section B: 20

Total number of simple graphs that can be drawn using six vertices are:

A. 2^{15} B. 2^{14} C. 2^{13} D. 2^{12}

nielit2017dec-assistanta discrete-mathematics graph-theory

Answer key

9.0.7 NIELIT 2016 MAR Scientist C - Section B: 20

Degree of each vertex in K_n is

A. n B. $n-1$ C. $n-2$ D. $2n-1$

nielit2016mar-scientistc discrete-mathematics graph-theory

Answer key

9.0.8 NIELIT 2021 Dec Scientist A - Section B: 60

If R and D are the radius and diameter of the graph $\mathrm{K}_{4,7}$, then the ordered pair (R, D) is equal to :

A. $(2,2)$ B. $(1,2)$ C. $(2,4)$ D. $(1,3)$

nielit2021dec-scientista graph-theory

Answer key

9.0.9 NIELIT 2021 Dec Scientist A - Section B: 53

The largest number of faces in a simple connected maximal planar graph with 100 vertices is :

A. 200 B. 198 C. 196 D. 96

nielit2021dec-scientista graph-theory

Answer key☟

9.0.10 NIELIT 2022 April Scientist B | Section B | Question: 104

A connected planar graph having 6 vertices, 7 edges contains ________ regions.

A. 15 B. 11 C. 3 D. 1

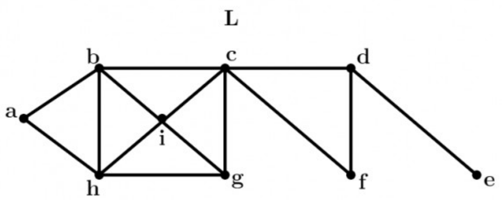
nielit2022apr-scientistb graph-theory

Answer key☟

9.1 Bridges (2)

9.1.1 Bridges: NIELIT 2017 July Scientist B (CS) - Section B: 11

Consider the following graph L and find the bridges,if any.



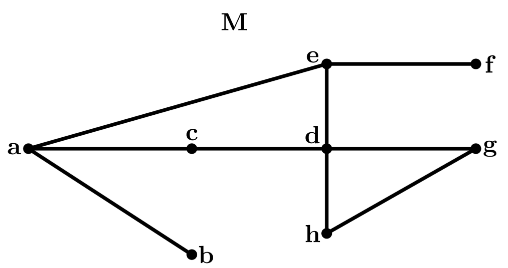
A. No bridge B. $\{d, e\}$ C. $\{c, d\}$ D. $\{c, d\}$ and $\{c, f\}$

nielit2017july-scientistb-cs discrete-mathematics graph-theory bridges

Answer key☟

9.1.2 Bridges: NIELIT 2017 July Scientist B (IT) - Section B: 6

Considering the following graph, which one of the following set of edges represents all the bridges of the given graph?



A. $(a, b), (e, f)$ B. $(a, b), (a, c)$ C. $(c, d), (d, h)$ D. (a, b)

nielit2017july-scientistb-it discrete-mathematics graph-theory bridges

Answer key☟

9.2 Counting (1)

9.2.1 Counting: NIELIT 2021 Dec Scientist B - Section B: 1

The number of un-labeled non-isomorphic graphs with four vertices is:

A. 12 B. 11 C. 10 D. 9

nielit-2021-it-dec-scientistb graph-theory counting

Answer key☟

9.3 Cycle (1)

9.3.1 Cycle: NIELIT 2017 DEC Scientist B - Section B: 23

Let G be a complete undirected graph on 8 vertices. If vertices of G are labelled, then the number of distinct cycles of length 5 in G is equal to:

A. 15 B. 30 C. 56 D. 60

nielit2017dec-scientistb discrete-mathematics graph-theory cycle

Answer key☟

9.4 Degree of Graph (7)

9.4.1 Degree of Graph: NIELIT 2016 MAR Scientist B - Section B: 3

Maximum degree of any node in a simple graph with n vertices is

A. $n-1$ B. n C. $n/2$ D. $n-2$

nielit2016mar-scientistb discrete-mathematics graph-theory degree-of-graph

Answer key☟

9.4.2 Degree of Graph: NIELIT 2017 July Scientist B (CS) - Section B: 2

Which of the following statements is/are TRUE for an undirected graph?

P. Number of odd degree vertices is even
Q. Sum of degrees of all vertices is even

A. P only
B. Q only
C. Both P and Q
D. Neither P nor Q

nielit2017july-scientistb-cs discrete-mathematics graph-theory degree-of-graph

9.4.3 Degree of Graph: NIELIT 2017 July Scientist B (IT) - Section B: 1

Given an undirected graph G with V vertices and E edges, the sum of the degrees of all vertices is

A. E B. $2E$ C. V D. $2V$

nielit2017july-scientistb-it discrete-mathematics graph-theory degree-of-graph

Answer key☟

9.4.4 Degree of Graph: NIELIT 2017 OCT Scientific Assistant A (CS) - Section B: 7

The number of the edges in a regular graph of degree $'d'$ and $'n'$ vertices is

A. Maximum of n, d B. $n + d$ C. nd D. $nd/2$

nielit2017oct-assistanta-cs discrete-mathematics graph-theory degree-of-graph

Answer key☟

9.4.5 Degree of Graph: NIELIT 2021 Dec Scientist B - Section B: 13

For which one of the following sequences CAN NOT be a degree sequence of a graph of order 5?

A. $3, 3, 2, 2, 2$
B. $3, 3, 3, 3, 2$
C. $3, 3, 3, 2, 2$
D. $4, 3, 3, 2, 2$

nielit-2021-it-dec-scientistb graph-theory degree-of-graph

Answer key☟

9.4.6 Degree of Graph: NIELIT 2021 Dec Scientist B - Section B: 96

For which one of the following sequences CAN NOT be a degree sequence if a graph of order 5 ?

A. $3, 3, 2, 2, 2$ B. $3, 3, 3, 3, 2$ C. $3, 3, 3, 2, 2$ D. $4, 3, 3, 2, 2$

nielit2021dec-scientistb graph-theory degree-of-graph graph-algorithms discrete-mathematics

9.4.7 Degree of Graph: NIELIT Scientific Assistant A 2020 November: 96

In an undirected graph, if we add the degrees of all vertices, it is:

A. odd
B. even
C. cannot be determined
D. always $n + 1$, where n is number of nodes

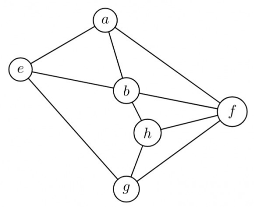
nielit-sta-2020 graph-theory easy degree-of-graph

Answer key☟

9.5 Depth First Search (1)

9.5.1 Depth First Search: NIELIT 2017 July Scientist B (IT) - Section B: 5

In a given following graph among the following sequences:



I. abeghf
II. abfehg
III. abfhge
IV. afghbe

Which are depth first traversals of the above graph?

A. I,II and IV only B. I and IV only C. II,III and IV only D. I,III and IV only

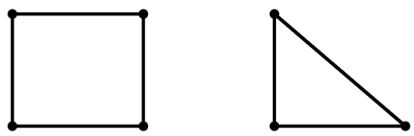
nielit2017july-scientistb-it discrete-mathematics graph-theory depth-first-search

Answer key

9.6 Euler Graph (2)

9.6.1 Euler Graph: NIELIT 2017 July Scientist B (CS) - Section B: 12

The following graph has no Euler circuit because



A. It has 7 vertices.
B. It is even-valent (all vertices have even valence).
C. It is not connected.
D. It does not have a Euler circuit.

nielit2017july-scientistb-cs discrete-mathematics graph-theory euler-graph

Answer key

9.6.2 Euler Graph: NIELIT 2017 July Scientist B (IT) - Section B: 7

Which of the following statements is/are TRUE?

$S1$:The existence of an Euler circuit implies that an Euler path exists.

$S2$:The existence of an Euler path implies that an Euler circuit exists.

A. $S1$ is true.
B. $S2$ is true.
C. $S1$ and $S2$ both are true.
D. $S1$ and $S2$ both are false.

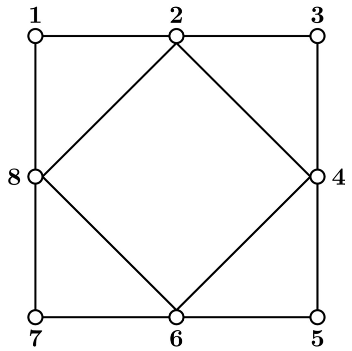
nielit2017july-scientistb-it discrete-mathematics graph-theory euler-graph

Answer key

9.7 Graph Algorithms (1)

9.7.1 Graph Algorithms: NIELIT 2021 Dec Scientist B - Section B: 73

What will be the Eulerian tour of the following graph G ?



G

A. $1\text{–}2\text{–}3\text{–}4\text{–}5\text{–}6\text{–}7\text{–}8\text{–}6\text{–}4\text{–}2\text{–}8\text{–}1$
B. $1\text{–}2\text{–}3\text{–}4\text{–}5\text{–}6\text{–}7\text{–}8$
C. $1\text{–}2\text{–}3\text{–}4\text{–}5\text{–}6\text{–}7\text{–}8\text{–}1$
D. $8\text{–}7\text{–}6\text{–}5\text{–}4\text{–}3\text{–}2\text{–}1$

nielit2021dec-scientistb graph-theory graph-algorithms

Answer key☟

9.8 Graph Coloring (1)

9.8.1 Graph Coloring: NIELIT 2017 DEC Scientist B - Section B: 52

Let G be a simple undirected graph on $n = 3x$ vertices $(x \geq 1)$ with chromatic number 3, then maximum number of edges in G is

A. $n(n-1)/2$ B. n^{n-2} C. nx D. n

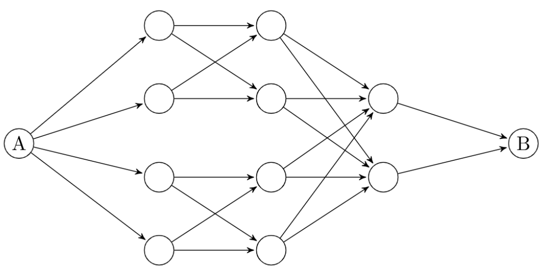
nielit2017dec-scientistb discrete-mathematics graph-theory graph-coloring

Answer key☟

9.9 Graph Connectivity (1)

9.9.1 Graph Connectivity: NIELIT Scientific Assistant A 2020 November: 19

What is the total number of ways to reach from A to B in the network given?



A. 12 B. 16 C. 20 D. 22

nielit-sta-2020 graph-theory graph-connectivity

Answer key☟

9.10 Graph Isomorphism (1)

9.10.1 Graph Isomorphism: NIELIT 2021 Dec Scientist B - Section B: 83

The number of un-labeled non-isomorphic graphs with four vertices is :

A. 12 B. 11 C. 10 D. 9

nielit2021dec-scientistb graph-theory combinatory graph-isomorphism

9.11 Graph Planarity (6)

9.11.1 Graph Planarity: NIELIT 2016 DEC Scientist B (CS) - Section B: 5

Let G be a simple undirected planar graph on 10 vertices with 15 edges. If G is a connected graph, then the number of bounded faces in any embedding of G on the plane is equal to:

A. 3 B. 4 C. 5 D. 6

nielit2016dec-scientistb-cs discrete-mathematics graph-theory graph-planarity

Answer key☟

9.11.2 Graph Planarity: NIELIT 2017 DEC Scientific Assistant A - Section B: 38

If a planner graph, having 25 vertices divides the plane into 17 different regions. Then how many edges are used to connect the vertices in this graph.

A. 20 B. 30 C. 40 D. 50

nielit2017dec-assistanta discrete-mathematics graph-theory graph-planarity

Answer key☟

9.11.3 Graph Planarity: NIELIT 2017 July Scientist B (CS) - Section B: 14

If G is an undirected planar graph on n vertices with e edges then

A. $e \leq n$ B. $e \leq 2n$ C. $e \leq 3n$ D. None of the option

nielit2017july-scientistb-cs discrete-mathematics graph-theory graph-planarity

Answer key☟

9.11.4 Graph Planarity: NIELIT 2017 July Scientist B (CS) - Section B: 15

Choose the most appropriate definition of plane graph.

A. A simple graph which is isomorphic to hamiltonian graph.

B. A graph drawn in a plane in such a way that if the vertex set of graph can be partitioned into two non-empty disjoint subset X and Y in such a way that each edge of G has one end in X and one end in Y.
C. A graph drawn in a plane in such a way that any pair of edges meet only at their end vertices.
D. None of the option.

nielit2017july-scientistb-cs discrete-mathematics graph-theory graph-planarity

Answer key☟

9.11.5 Graph Planarity: NIELIT 2017 July Scientist B (IT) - Section B: 12

Let G be a simple connected planar graph with 13 vertices and 19 edges. Then, the number of faces in the planar embedding of the graph is

A. 6 B. 8 C. 9 D. 13

nielit2017july-scientistb-it discrete-mathematics graph-theory graph-planarity

Answer key☟

9.11.6 Graph Planarity: NIELIT 2017 July Scientist B (IT) - Section B: 8

A connected planar graph divides the plane into a number of regions. If the graph has eight vertices and these are linked by 13 edges, then the number of regions is:

A. 5 B. 6 C. 7 D. 8

nielit2017july-scientistb-it discrete-mathematics graph-theory graph-planarity

Answer key☟

Answer Keys

9.0.1	C	9.0.2	C	9.0.3	B	9.0.4	D	9.0.5	B
9.0.6	A	9.0.7	B	9.0.8	A	9.0.9	C	9.0.10	C
9.1.1	B	9.1.2	A	9.2.1	B	9.3.1	C	9.4.1	A
9.4.2	C	9.4.3	B	9.4.4	D	9.4.5	C	9.4.6	C
9.4.7	B	9.5.1	D	9.6.1	C	9.6.2	A	9.7.1	A
9.8.1	C	9.9.1	B	9.10.1	B	9.11.1	D	9.11.2	C
9.11.3	B	9.11.4	C	9.11.5	B	9.11.6	C		